Impact of Grip Distance on Muscle Activation in the Chest, Shoulders, and Triceps During the Bench Press

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2.671 Measurement and Instrumentation

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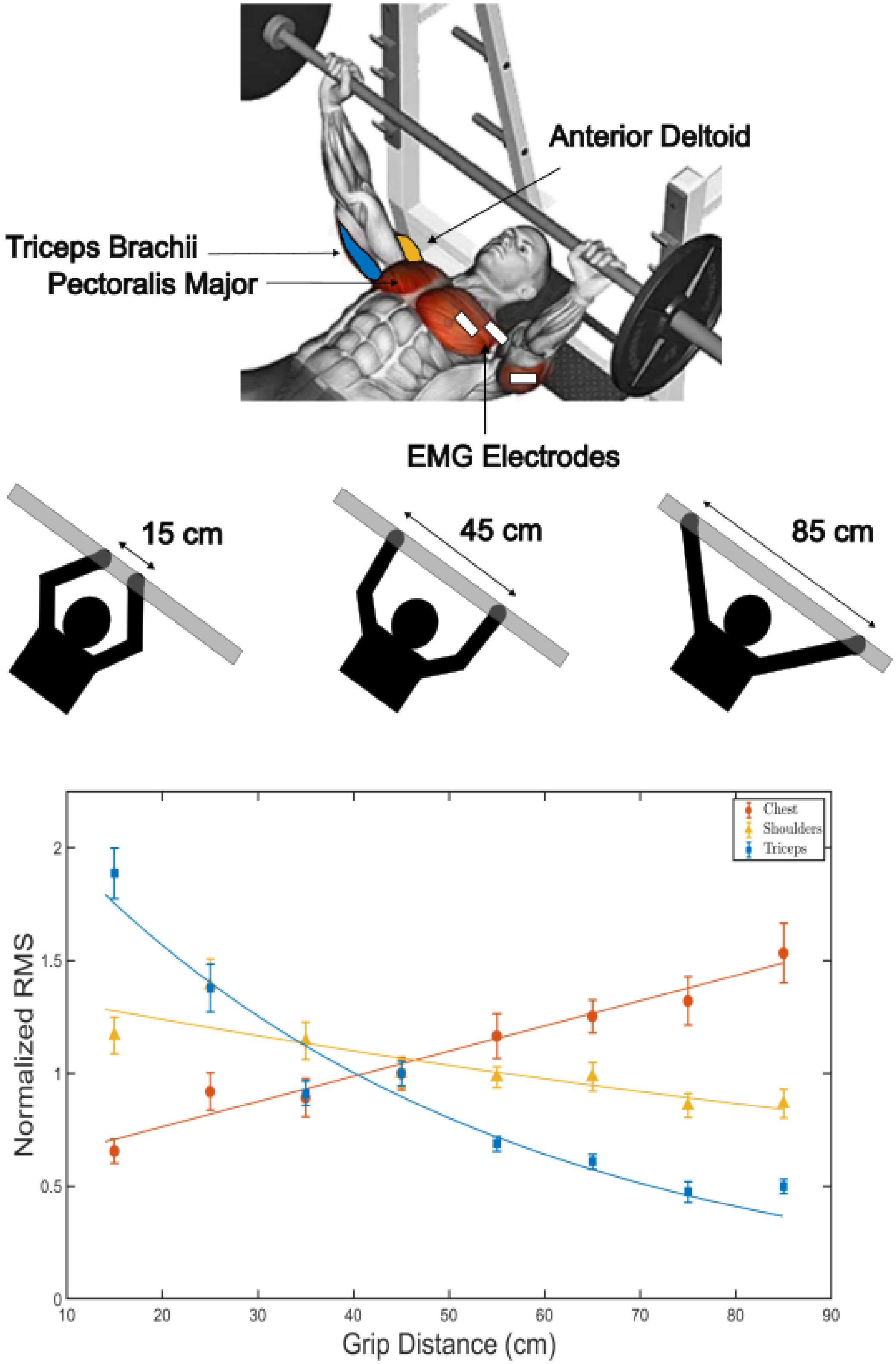
Visual Abstract



Abstract

Understanding how grip distance on the bench press affects muscle activation in the chest, shoulders, and triceps allows weightlifters to target specific muscle groups, as higher activation correlates with greater strength improvement. In this study, surface electromyography sensors were attached to these muscles to determine the activation, and an experienced lifter performed 30 repetitions with a 15 kg barbell across eight evenly spaced grip distances ranging from 15 cm to 85 cm. The activation profiles for each muscle group were determined during repetitions at each grip distance. Widening the grip distance from 15 cm to 85 cm resulted in a 133% increase in chest muscle activation, accompanied by a 25% decrease in shoulders and a 73% decrease in the triceps. A wider grip distance is beneficial for chest development, whereas a narrower grip distance is effective for developing the shoulders and triceps.

Keywords: Grip Distance, Bench Press, EMG, Muscles

1. Introduction

The bench press is one of the most popular exercises for developing overall upper-body strength because it is a compound movement that primarily engages the pectoralis major (chest), anterior deltoid (shoulders), and triceps brachii (triceps) [1]. During the bench press, the barbell is typically held slightly wider than the biacromial distance, the shoulder width, and lowered to the chest and back up until the elbows are fully extended. Variations of the exercise increases muscle activation in a separate part of the body by utilizing a wider or narrower grip distance. Its adjustable hand positioning allows weightlifters to target specific muscle groups more effectively by reorienting the path of the arm movement, shifting the emphasis on the muscles during each repetition [2]. Understanding how grip distance affects muscle activation will provide valuable insights for weightlifters to optimize their training.

In this study, electromyography (EMG) analysis was implemented to determine the muscle activities in the chest, shoulders, and triceps. Surface EMG (sEMG) electrodes were attached to the chest, shoulders, and triceps while an experienced lifter bench-pressed a 15 kg barbell with varying grip distances. The initial hand separation was 15 cm and increased by 10 cm each trial until the grip distance reached 95 cm. Raw data collected from the sensors were compiled and analyzed to produce a fit model that illustrated how grip distance affected muscle activation. The measurements were normalized by dividing the data by a reference EMG signal collected on the corresponding day of data collection. The normalized roots mean square (RMS) values from each muscle group were plotted against grip distance to help weightlifters visually identify the relative activations to choose the optimal grip width for their training.

2. Background

2.1 Building Muscles

An individual can build strength by repeatedly engaging their muscles under load, which stimulates hypertrophy, the enlargement of muscle fibers resulting from resistance training. The three primary factors responsible for hypertrophic responses are mechanical tension, muscle damage, and metabolic stress [3]. Mechanical tension involves the stretching and contracting of muscle fibers, which increases their length. Muscle damage refers to the microscopic tears in the fibers. When the body repairs those tears, the muscles thicken and become more resistant to future tears. Metabolic stress refers to the body's chemical response during resistance training, where normal cellular processes are disrupted and byproducts accumulate, triggering growth-oriented body responses. Through repeated cycles of tears, repair, and growth, the muscles gradually adapt to the constant stress experienced by the body, increasing muscle mass and strength. During the bench press, changing the grip distance to target a specific muscle would lead to a greater hypertrophic response in that muscle.

Measuring the effort made by the muscles during a lift is determined by the amount of muscle activation. The activation comes from excitation-contraction coupling, where the brain sends electrical signals to the muscles, initiating contraction and generating force [3]. Lifting heavier weights require greater muscle engagement, which corresponds to higher muscle activation compared to lifting lighter weights. Muscle activation plays a key role in strength development by allowing the muscles to adapt to external loads. During a bench press, the muscle group generating the greatest force will produce the strongest electrical signals, and vice versa, which are detected by the EMG sensors. Knowing the relative activation of each muscle helps determine the grip distance that best promotes strength development for each muscle group. Figure 1 shows the three primary muscles engaged during the bench press.

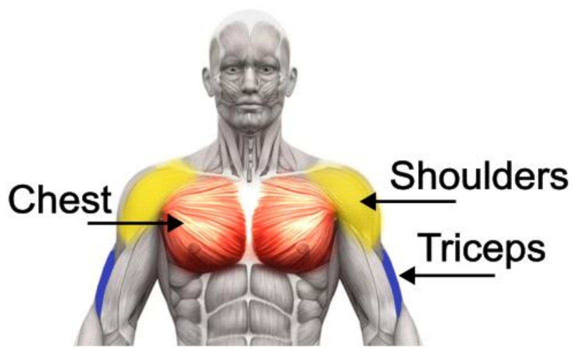


Figure 1: The highlighted regions illustrate the three primary muscle groups activated during the bench press: the chest, shoulders, and triceps [4]. Varying the grip distance on the bench press changes the activation level of each muscle group.

2.2 Capturing and Analyzing EMG Signals

Electromyography is a diagnostic test that measures signals produced by action potential across tissues [5]. sEMG electrodes are placed on the skin to record the EMG signals in the muscle, as shown in Figure 2. The reliability of these electrodes is evaluated using the intraclass correlation coefficient, which compares measurements across electrodes to determine how similar the results are, with values ranging from 0 to 1. sEMG is considered unreliable when the reliability coefficient is below 0.5, moderate between 0.5 and 0.75, good between 0.75 and 0.9, and excellent when greater than 0.9 [6]. Seven studies have reported reliability coefficients between 0.66-0.99 [6], suggesting that sEMG are generally reliable and capable of reproducing data. Reliable signals retrieved by the electrodes are represented on a potential versus time graph, providing information about the amplitude of the EMG signals.

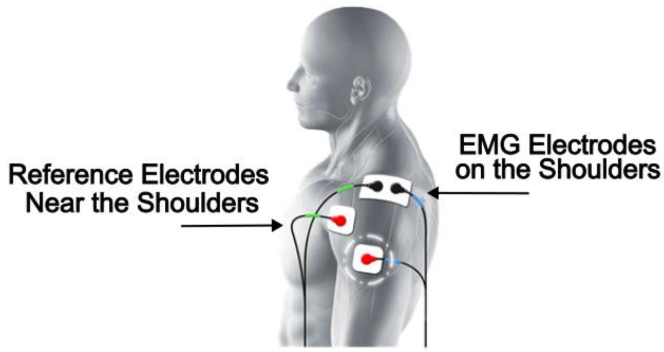


Figure 2: sEMG electrodes are placed on the surface of the subject's shoulders, and the reference electrodes are placed near the shoulders [7]. The EMG and reference electrodes enable electrical signals to be measured across the muscle.

In EMG analysis, the signals are converted into RMS values, which reflect the strength of the signal within a given time window. These RMS values were then normalized by dividing them by the mean RMS of a baseline motion, providing a reference for each data point. This normalization enables comparable data throughout multiple trials by limiting the variability caused by the sensors working differently across different days and place on the body.

2.3 Biomechanics of the Bench Press

The contribution of the chest, shoulders, and triceps during the bench press depends on the range of motion (ROM) at the shoulder and elbow joints and the respective moment arms. ROM is the change in joint angles, and the moment arm is the horizontal distance between where the weight is held and the joint. A larger ROM increases muscle activation by engaging a greater number of muscle fibers. Likewise, a longer moment arm increases activation because the muscle requires a greater torque to lift the barbell, since torque is proportional to the moment arm. Since these two factors vary with grip distance, the activation profile of the chest is different from that of the shoulders and triceps, so a separate fitting model is used.

During the bench press, the chest generates most of the torque required to move the barbell, so a linear fit model is applied to the data. Meanwhile, the shoulders and triceps assist the chest in lifting the barbell, so the ROM is the dominant factor in determining muscle activation. Since their ROM is limited by the anatomical structure of the joints, their muscle activation will reach an asymptote. Therefore, an exponential fit model is used for the shoulders and triceps. As shown later, these models are consistent with experimental data.

2.4 Current Knowledge

When exploring the biomechanics of the body, increasing the grip distance during a bench press increases the moment arm of the shoulders, resulting in higher EMG signals in the chest [8]. Conversely, decreasing the grip distance increases the moment arms of the elbows, leading to greater EMG signals in the shoulders and triceps. Although biomechanics offers a framework for predicting these outcomes, current studies suggest inconclusive relationships between muscle activity and grip distances. The relationship remains unclear because the findings from existing studies are contradictory. For example, no significant differences in muscle activation across a narrow, medium, and wide grip were reported by Saeterbakken et al. [9]. On the contrary, a wider grip width resulted in greater muscle activation in the chest and lower shoulders, and reduced triceps activation, as reported by Barnett et al. [10]. These findings contradict each other and do not provide a clear relationship between muscle activation and grip distance.

In the present study, using a variable grip distance rather than discrete categories enables data collection between the narrow, medium, and wide grips. The data is used to plot a fit model of normalized RMS (nRMS) as a function of grip distance, including grip distances similar to the narrow, medium, and wide positions defined in previous studies. By visualizing activation trends for the chest, shoulders, and triceps, the results could be compared with the biomechanical model to evaluate their agreement with previous studies.

3. Experimental Design

To determine the EMG signals in the chest, shoulders, and triceps during the bench press, the subject repeated multiple repetitions of the bench press at varying grip distances.

3.1 Bench Press Measurement Setup

A masking tape marked at multiple grip distances, starting at 15 cm and increasing by 10 cm for each trial up to 95 cm, was attached to a 15 kg barbell for the bench press. GS-27 disposable electrodes were attached to the middle of the chest, shoulders, and triceps, with the reference electrode placed away from the muscle of interest. The sEMG electrodes were connected to the Motion Lab 2-channel EMG sensors, which detected electrical impulse during muscle contraction and amplified the signals into the Vernier LabQuest Mini. An Interlink 406 Force Sensing Resistor (FSR) Force Pad was taped onto the palm of each hand to measure the force exerted on the hand when bench pressing. The FSR detected changes in the resistance based on the applied force and outputs a voltage via voltage divider. When there was no force, the circuit remained open and output approximately 0 volts. The Force pads were connected to a separate Vernier LabQuest Mini, and both LabQuest Minis were connected to a laptop running Logger Pro for EMG and force data collection. A diagram of the experimental setup is shown in Figure 3.

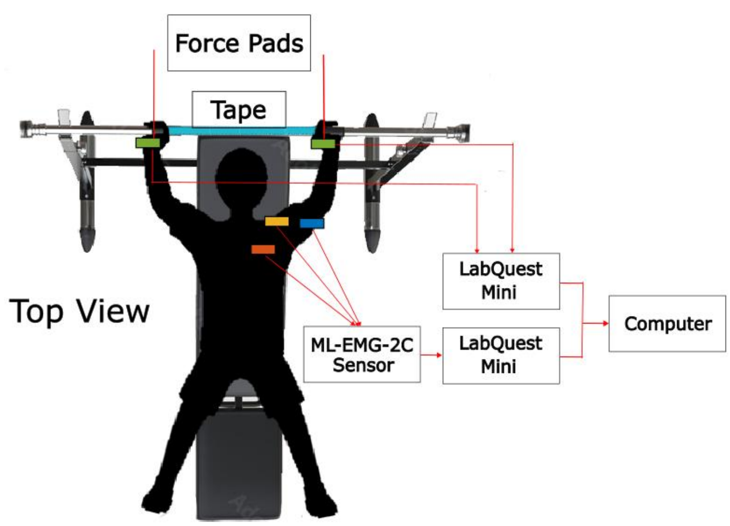


Figure 3: The experimental setup where the subject has force pads attached to their palms and sEMG electrodes placed on the chest, shoulders, and triceps. Grip distances were determined by reading the marked distances on the tape. The setup

made it possible to capture EMG signals from each muscle at the specified grip distances.

3.2 Performing Repetitions

Each day, before collecting the main data, a normalization trial was performed at a 45 cm grip distance using the same procedures as a regular trial. EMG signals were then recorded for eight grip distances, starting at 15 cm and increasing by 10 cm for each trial up to 95 cm. For each grip distance, the subject performed 30 repetitions, and each repetition is defined as lowering the barbell to the chest and pressing up until the elbows were fully extended, as shown in Figure 4. The EMG signals were sampled at 2 kHz, twice the recommended rate for typical EMG signals [11]. A five-minute rest period was provided between each set to allow for adequate recovery. Together, these procedures ensured reliable and comparable EMG measurements across multiple days.

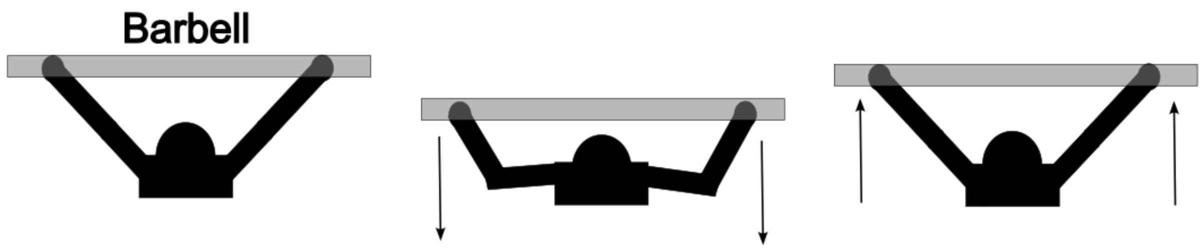


Figure 4: A repetition on the bench press is when the subject lowers the barbell down to their chest and then pushes it back up until the elbows are fully extended. A consistent standard is established for how repetitions are performed.

4. Results and Discussion

4.1 Root Mean Square and Normalization

For each grip distance, the force pad operated in parallel with the EMG sensors to determine when a repetition was performed. The force sensor recorded dips, as shown in Figure 5a, that represented the movement of the barbell being lowered to the chest and raised back up. There were 30 dips corresponding to 30 repetitions, and each dip contained drops because the net acceleration of the barbell decreased after each repetition. The segment of the EMG signal defining a repetition was determined by its respective time interval. After subtracting the mean to eliminate the offset caused by holding the barbell, the remaining signal was converted into its root mean square (RMS) value, as illustrated in Figure 5b.

(a)

(b)

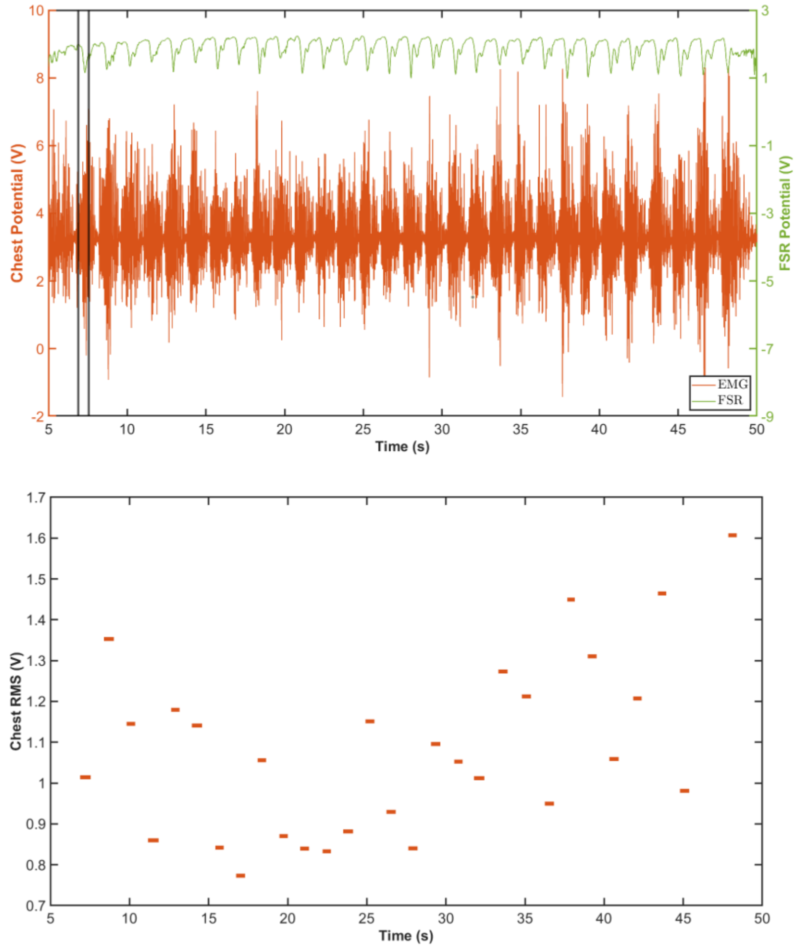


Figure 5: (a) EMG signals for the chest and FRS readings at a grip distance of 75 cm. The first repetition occurred when the FSR recorded a dip, corresponding to the time interval 6.88-7.55 s. (b). Plot showing the chest RMS values for each repetition at the same grip distance after the time intervals were identified. The RMS was calculated after subtracting the mean, and the first repetition yielded an RMS value of 1.014 V.

The RMS values for the chest, shoulders, and triceps were determined for each repetition at every grip distance, including the data for reference activation. Before plotting RMS as a function of grip distance, the data were normalized by dividing all RMS values from the main trials by the mean RMS over the entire duration of the reference data collected on the same day. Figure 6 displays the mean RMS values with error bar along with their fit curves. Table 1 lists the parameters of the corresponding fit curves along with the type of fit model used in Figure 6.

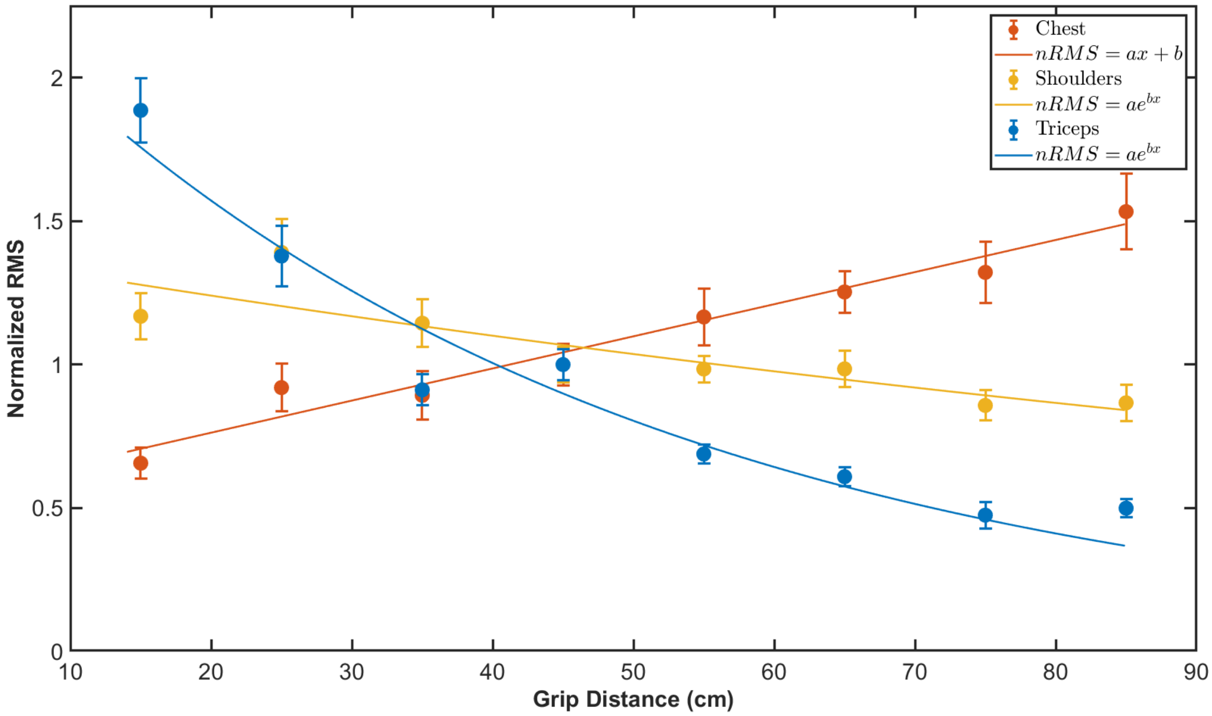


Figure 6: Mean values and error bars for each muscle group at each grip distance. Curve fits are applied for each muscle group with the equations and parameters listed in Table 1.

Table 1: Fit equations and parameters used in Figure 6.

	Fit Equation	a	b
Chest	nRMS = $ax + b$	$(1.12 \pm 0.14) \times 10^{-2} \frac{1}{cm}$	$(5.34 \pm 0.77) \times 10^{-1}$
Shoulders	nRMS = ae^{bx}	1.402 ± 0.080	$(-6.0 \pm 1.1) \times 10^{-3} \frac{1}{cm}$
Triceps	nRMS = ae^{bx}	2.48 ± 0.13	$(-2.23 \pm 0.15) \times 10^{-2} \frac{1}{cm}$

By visually inspecting Figure 6, muscle activation in the chest varies inversely with that of the shoulders and triceps as grip distance changes on the bench press. As grip distance increases, the chest exhibits a positive trend, while the shoulders and triceps display negative trends. At a narrow grip distance of 15cm, the nRMS is 0.657 ± 0.055 for the chest, 1.168 ± 0.080 for the shoulders, and 1.87 ± 0.11 for the triceps. At a wider grip distance of 85 cm, the nRMS is 1.53 ± 0.13 for the chest, 0.867 ± 0.064 for the shoulders, and 0.500 ± 0.032 for the triceps. The results show a trade-off in muscle activation across different grip distances. To emphasize chest activation, a wider grip distance should be used. On the contrary, greater engagement of the shoulders and triceps is achieved using a narrower grip distance. A balance in muscle activation across all three muscle groups can be attained using a grip distance between 35 cm to 55 cm. These outcomes are specific to the subject's height of 163 cm.

4.2 Biomechanical Reasonings

Figure 6 can be explained by the biomechanics of the body, particularly the ROM and the moment arms of the shoulders and elbows. ROM and moment arms of the subject were calculated manually, and for simplicity, the chest and shoulders shared identical values. The subject's position with the barbell is further reduced to a six-sided polygon, as illustrated in Figure 7a. The forearm length, upper arm length, grip distance, and shoulder width form the six sides of the polygon, with measurements shown in table 2. Since the barbell touches the chest during the repetition, the final height of the polygon is known, allowing calculation of the initial and final joint angles. The initial and final angle of the shoulders and elbows were calculated using the law of cosine (1). Figure 7b highlights the ROM and moment arm plots for the chest, shoulders, and triceps across the different grip distances.

$$C^2 = A^2 + B^2 - 2ABcos(\theta_c) \quad (1)$$

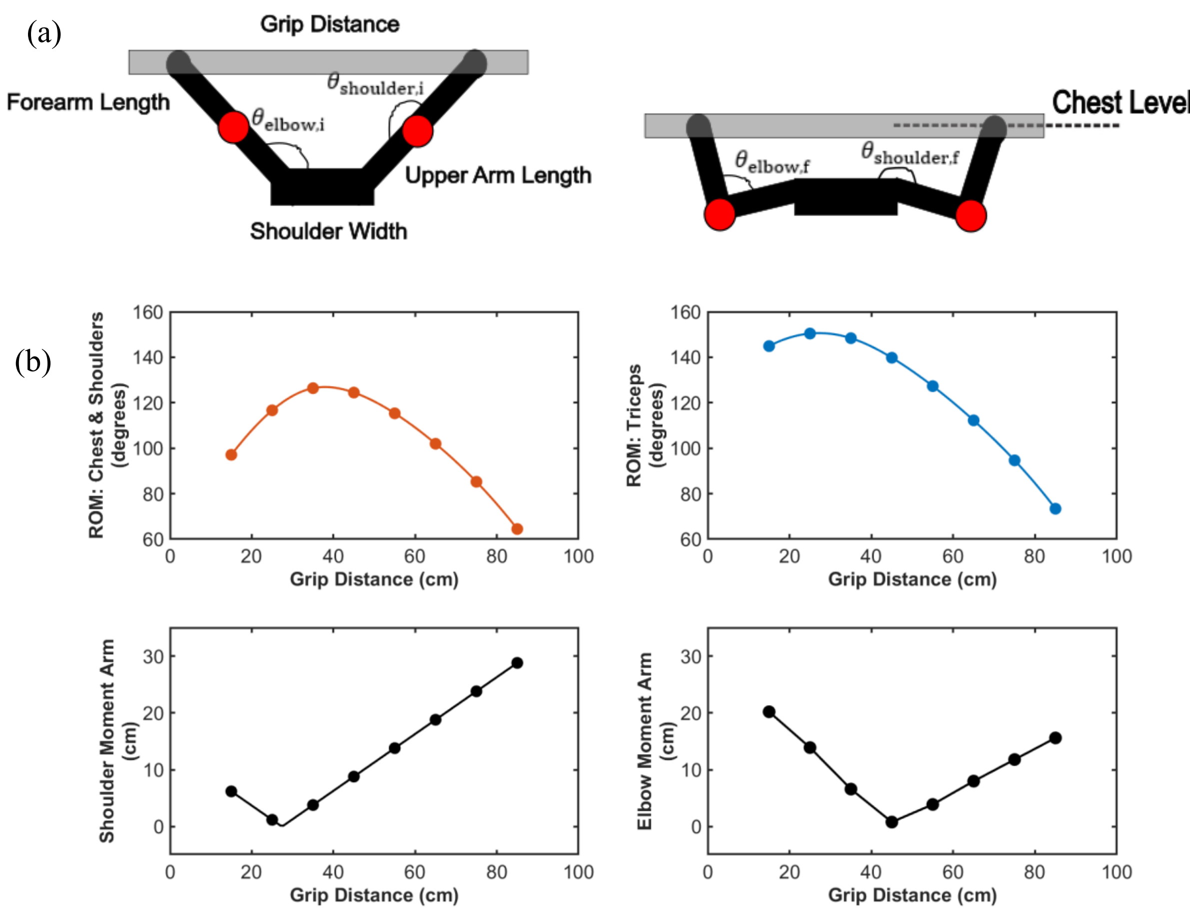


Figure 7: (a) Diagram of the change in the angle of the shoulders and elbows at the beginning of a repetition to when the barbell reaches the chest. The difference in angles is the range of motion. (b) Scatterplot of the ROM and moment arm for each muscle group at the corresponding grip distance. Shoulder and elbow moment arms are plotted in black, chest and shoulder ROM in orange, and triceps ROM in blue.

Table 2: Measurements taken to produce the results in Figure 7b.

	Values (cm)
Forearm Length	24.7 ± 1.5
Upper Arm Length	14.5 ± 1.0
Shoulder Width	35.4 ± 3.5
Final Height	14 ± 1.5

The experiment produced results and trends consistent with those reported by Mausehund et al., where increasing the grip distance resulted in higher EMG signals in the chest, while decreasing the grip distance increased the EMG signals in the shoulders and triceps [8]. In Figure 7b, smaller grip distances enabled a greater ROM in the chest, resulting in higher EMG signals. Meanwhile, the moment arms for the shoulders and triceps decreased, so lower EMG signals were observed. With wider grip distances, the moment arm with respect to the chest increased, producing a positive trend in EMG signals. The ROM of the shoulders and elbows decreased, resulting in smaller EMG signals. Figure 8 provides a graphical representation of how EMG signals change with varying ROM and moment arms. Figure 8a and 8c are the fit models illustrated in Figure 6, and Figure 8b and 8d are normalized RMS values based on the biomechanics of the body.

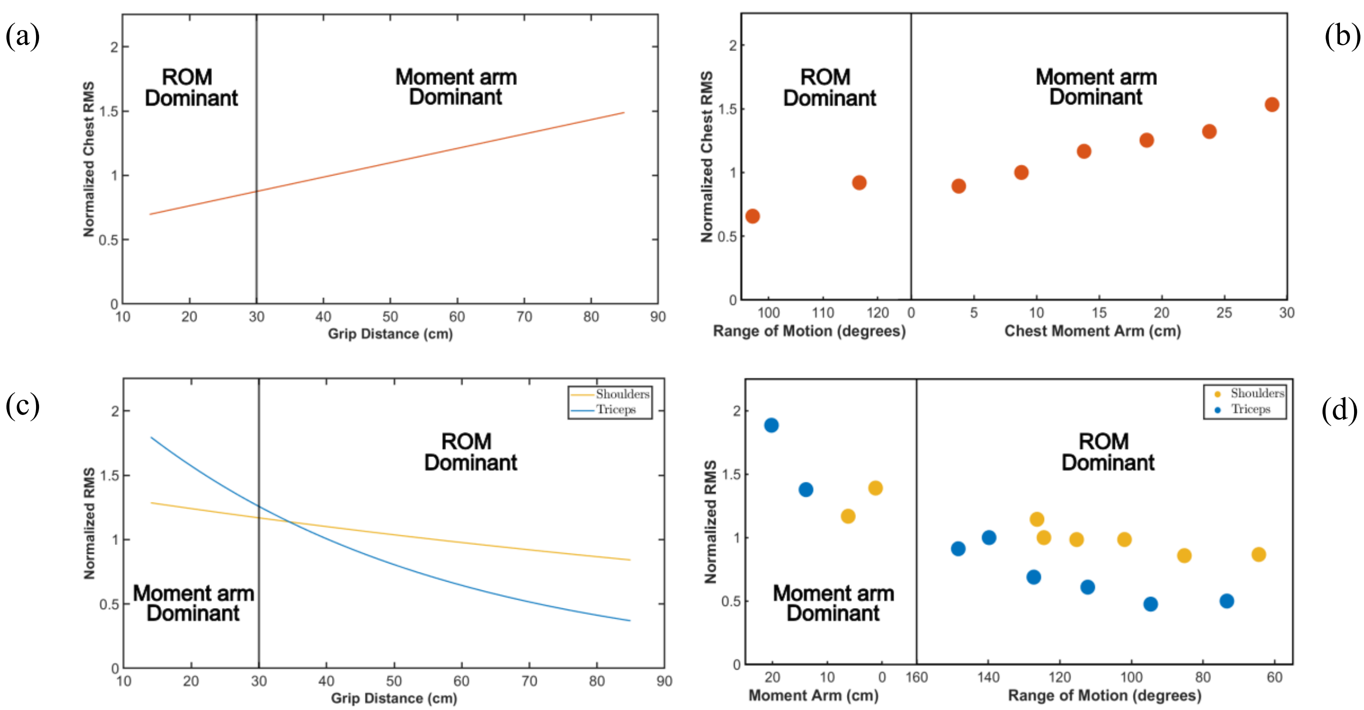


Figure 8: (a) Linear fit of the chest muscle activation divided into regions where ROM and moment arm heavily influence the activation. (b) Plot of nRMS for chest in terms of ROM and its moment arm. (c) Exponential fit of the shoulders and triceps muscle activation is divided into regions where ROM and moment arm heavily influence the activation (d). Plot of nRMS for the shoulders and triceps in terms of ROM and their moment arms. Together, the plots show similar activation trends that align with the biomechanics of the body.

The trends in Figure 8 are consistent with those presented in Figure 6, where the chest displays an increase in EMG signals while the shoulders and triceps show a decrease. In Figure 7b, the simplified biomechanical model shows the chest and shoulders sharing the same ROM. However, Figures 8c and 8d compare the shoulders and triceps on the same plot because both muscles assist the chest in lifting the barbell, and ROM becomes the primary factor influencing muscle activation at wider grip distances. Since the biomechanical analysis aligns with both the experimental results and the findings of Mausehund et al., the results indicate that changing the grip distance on the bench press changes the ROM and moment arms, which in turn influence the EMG signals produced by the muscles.

5. Conclusion

The experimental results reveal that changing the grip distance on the bench press has a significant effect on muscle activation in the chest, shoulders, and triceps. As grip distance increases, chest activation rises while shoulders and triceps decrease. These trends are corroborated by human biomechanical patterns and agree with the findings reported by Mausehund et al. Weightlifters can use Figure 6 to visually determine the grip distance needed to emphasize specific muscles. To target chest development, a wider grip distance is recommended, whereas narrower grips are more effective for training the shoulders and triceps.

Although the experiment demonstrated expected results that align with the biomechanical perspective, factors such as normalization method and fatigue could be further studied to improve the results. Future studies could also use goniometers to directly measure range of motion, rather than calculating it manually, allowing for a more accurate biomechanical model.

Acknowledgement

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